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RESEARCH ARTICLE

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The effects of unilateral muscle fatigue on bilateral physiological tremor

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Abstract The aim of this study was to examine the post-exercise effects of fatiguing the wrist extensor muscles of a single arm on postural tremor and muscle activity in both arms. Previous research has shown that, for neurologically normal subjects, the tremor seen within a single limb segment is uncorrelated to that seen contralaterally. However it has been speculated that some bilateral relation does exist, and that the nature of the relation may only become evident under conditions where the neuromuscular system is perturbed. To further investigate this potential bilateral relation, seven healthy subjects were required to adopt a bilateral postural pointing position after exercise-induced fatigue of the wrist extensor muscles of a single arm. Tremor from the forearm, hand and finger segments of each arm, surface EMG activity from extensor digitorum (ED) of each arm, and blood lactate data were collected prior to and after the exercise intervention. The main result was that fatiguing the distal muscles of one arm resulted in a bilateral increase in both the physiological tremor and ED activity. The change in tremor was confined to the index finger with no change in the tremor for the hand or forearm segments of either arm. While three peaks were seen in the frequency profile of the finger tremor, the effects of fatigue were confined to an increase in the peak power of the neurally generated 8–12 Hz tremor component. The contralateral increase in muscle activity was also reflected by a change in the frequency profile of the EMG output, with an increase in the peak power of both muscles following exercise of the wrist extensors of a single arm. The bilateral increases in physiological tremor and EMG activity of ED were only observed during the bilateral pointing task, with no changes in tremor or EMG activity seen for the non-exercised limb during the unilateral exercise protocol. The specificity of the resultant increases in the neurally generated 8–12 Hz component of finger tremor amplitude and EMG activity, coupled with the lack of any changes in tremor for the more proximal arm segments, indicate that these bilateral effects were mediated by an increase in the central neural drive to both limbs. Together this set of results challenges the general assumption of bilateral independence of tremor production, and further illustrate the task dependent nature of exercise-induced fatigue.

Keywords 8–12 Hz tremor · Fatigue · Bilateral · EMG · CNS

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Introduction

Fatigue has a major impact on motor function, altering the output of both the peripheral and central components of the neuromuscular system (Enoka 1995; Cresswell and Loscher 2000). It is apparent that fatigue is responsible for a number of changes seen in neuromuscular function, such as a decline in muscle force production, increased variability in isometric force output, and increased amplitude of postural tremor (Bousfield 1932; Lippold 1981; Palmer 1991; Enoka and Stuart 1992; Enoka 1995; Arihara and Sakamoto 1999; Ebenbichler et al. 2000; Gandevia 2001). With regard to physiological tremor, exercise-induced fatigue results in explicit changes in the time and frequency characteristics of the tremor output. The greatest effect being increased peak amplitude of the neurally generated, 8–12 Hz frequency component (Lippold 1981; Palmer 1991; Viitasalo and Gajewski 1994; Viitasalo et al. 1994; Saxton et al. 1995). Thus an intervention such as fatigue, which

directly alters the neurally generated component of tremor, has been employed to provide insight into the output of those neural sources underlying tremorgenesis (Lippold 1970; Llinas 1984; Llinas 1991).

Although a major component of tremor in a single limb segment is neurally derived, no studies have been able to demonstrate any significant degree of tremor coupling between limbs in neurologically normal subjects (Marsden et al. 1969; Lippold 1981; Arblaster and Lakie 1990; Arblaster et al. 1990; Morrison and Newell 1996). This lack of any significant contralateral tremor relations has prompted the suggestion that the neural component of physiological tremor is generated by parallel oscillators within the CNS (Marsden et al. 1969; Stein and Lee 1981). However, Morrison and Newell (1999) have shown that when a single arm is externally braced to restrict joint motion, there is a resultant change in the tremor amplitude in both index fingers and the hand-finger tremor coupling relations within each limb. These contralateral effects are seen even though no increase in the tremor of the more proximal arm segments (hand, forearm, upper arm) is seen nor is there any increase in the level of coupling between limbs. These results indicate that rather than the tremor in each limb being completely independent, some form of inter-limb tremor relation may exist, and that this link is probably neurally mediated. One suggestion is that the nature of any underlying common between-limb tremor relations would only emerge under conditions where a single limb or segment is perturbed in some manner (Morrison and Newell 1999). In this regard, any intervention that would result in increased neural drive to a single limb segment, and so increase the amplitude of the 8–12 Hz tremor component, may also produce significant increases in the 8–12 Hz tremor in the contralateral limb.

Previous research has documented the effects fatigue has on both tremor within a single limb (Lippold 1981; Palmer 1991; Viitasalo and Gajewski 1994; Viitasalo et al. 1994; Saxton et al. 1995) and contralateral EMG activity during tasks requiring high force output (Zijdewind et al. 1998; Zijdewind and Kernell 2001; Shinohara et al. 2003; Todd et al. 2003). However, the potential effects of unilateral fatigue on bilateral postural tremor and related EMG activity has not been extensively examined. Such an intervention may produce further insight as to the bilateral nature of tremor. For example, if unilateral fatigue induces bilateral effects due, in part, to increased neural activation, then it could be hypothesized that inducing fatigue within a single limb should alter the neural (8–12 Hz) component of tremor and related EMG activity in both limbs. Alternatively, if no common neural mechanism for contralateral tremor activity exists between limbs, then the effects of localized fatigue should only be manifested in the exercised limb.

The aim of this study was to examine the post-exercise effects of fatiguing the wrist extensor muscles of a single arm on postural tremor and muscle activity in both arms. Investigation of what changes occur in tremor profiles of limb segments located proximal and contralateral to the fatigued segment should provide further insight into the nature of the neural output underlying tremorgenesis, and the respective inter- and intra-limb tremor relations.

Methods

Subjects

Seven healthy university students (two male, five female, mean age 24 years, range 19–37 years) with no known neurological problems consented to participate in this study. All subjects were right-hand dominant, with little or no history of formal resistance training. Twelve hours prior to testing, subjects were required to abstain from any form of stimulant that may impact on limb tremor, including moderate-high intensity exercise, alcohol and/or caffeine ingestion. Written informed consent was obtained from each subject prior to testing. All experimental procedures complied with the guidelines of Griffith University Ethics Committee and were consistent with the Declaration of Helsinki.

Equipment and data collection

The effect of exercise-induced fatigue on bilateral tremor output and forearm muscle activity during a pointing task was assessed. Eight 30 s trials were performed where accelerometry, EMG, and blood lactate data were obtained for both pre- and post-exercise conditions.

Upper-limb tremor was measured pre- and post-exercise using six lightweight Coulbourn uniaxial accelerometers (V94–41, range ± 10 g). Accelerometers were attached to both upper limbs on the following landmarks: (1) forearm (over the brachioradialis muscle, approximately 6–7 cm distal to the elbow joint on the dorsal aspect), (2) hand (middle of third metacarpal), and (3) index finger (dorsal distal aspect), so that the measurement axis was perpendicular to the ground during bilateral pointing. Prior to testing, each accelerometer was calibrated by zero balancing in DC mode on a level surface. All accelerometer data were sampled at 100 Hz and amplified through a Coulbourn strain gage transducer coupler (V75–25A).

Bipolar Ag/AgCl EMG surface electrodes were positioned over the belly of the extensor digitorum (ED) muscle of both upper limbs to measure muscle activity during the performance of the postural pointing tasks pre- and post-exercise. Electrodes were placed with an inter-electrode distance of 2 cm, and in parallel with the direction of the underlying muscle fibres. All EMG signals were sampled at 1,000 Hz and amplified using Coubourn isolated bioamplifiers (V75–02). Impedance of the electrode-skin interface was kept below 5 kΩ.

Blood lactate levels were analysed using calibrated LactatePro portable lactate analysers (LT-1710, Arkray

Inc, Kyoto, Japan). Finger capillary blood was obtained from the middle digit of both limbs simultaneously at four discrete time periods: (1) at rest, prior to data collection, (2) immediately following pre-exercise trials, (3) immediately following exercise, and (4) immediately following post-exercise trials.

Experimental protocol

All subjects completed a postural pointing task prior to and following the exercise intervention. For the pointing task, subjects stood with both arms held parallel to the ground. In this position, the shoulder was flexed to 90° in the sagittal plane with the elbow fully extended, forearm pronated and the wrist held in the neutral position (with palm facing downwards). Both index fingers were extended at the metacarpophalangeal joint with the thumb adducted and fingers 3, 4 and 5 flexed to form a loose fist. The task goal for each condition, which was made explicit to each subject, was to focus on the tip of the index fingers and attempt to minimise the motion there.

MVC measurement At least 24 h prior to experimental testing, subjects attended a familiarization session where each participant's maximal voluntary contraction (MVC) for wrist extension was established. Subjects were randomly allocated to a preferred or non-preferred group, which designated the limb that they would use for the MVC, and subsequent fatiguing exercises. Limb preference was determined by asking the subjects which limb they preferentially used in tasks such as writing and throwing. For the purposes of this study, the limb used for the MVC and exercise will be referred to as the exercised limb.

Each subject performed the MVC contraction seated, with the upper body fully supported on the back-rest of a sturdy chair. Both upper limbs were flexed at the elbow (~90°) with each forearm resting on the chair armrest. For the exercised limb, the wrist was held in a pronated position. The forearm of the exercised limb was externally strapped to the armrest during the MVC in order to isolate the wrist extensor muscles and limit proximal joint movement. Subjects were instructed to maximally contract against a fixed load that was attached to a XTran 250 N S-beam load cell transducer (Applied Measurement). The MVC protocol consisted of 3 maximal wrist extension contractions of 5 s duration, with a 5-min rest between contractions. The greatest of the three trials was taken to represent each subject's MVC. The non-exercised limb remained in a relaxed position throughout the MVC testing. EMG activity of ED of the non-exercised limb was monitored during the MVC procedure to ensure the limb was relaxed.

Exercise-fatigue protocol Following the collection of pre-exercise tremor, EMG and lactate data, subjects performed a fatiguing task that consisted of full range wrist extension/flexion exercises using a predetermined dumbbell load set at 50% of MVC. Two exercise sets were performed to volitional fatigue with a 1-min rest period between sets. The average time the fatiguing exercise was performed was 61 s (±4). As with the MVC trials, subjects sat on the chair with the forearm of their exercised limb secured to the chair armrest. The trunk and upper limb posture adopted for the fatigue protocol was the same as stated for the MVC trials, however the wrist joint of the exercised limb was not fixed, allowing for full active range of motion. The total range of motion during each extension/flexion cycle corresponded to approximately 120° (wrist extension of ~60°, wrist flexion of ~ 60°). Following the exercise-fatiguing protocol, subjects once again adopted the bilateral pointing position, and post-exercise data were collected. Activity in the ED muscle of the non-exercised limb was monitored during the exercise protocol to assess whether the unilateral exercise resulted in any contralateral EMG changes. Subsequent analysis demonstrated that no significant change in EMG amplitude of the ED muscle for the non-exercised limb was seen during the exercise intervention.

Data analysis

The dependent variables assessed in the present study were derived from time and frequency domain analysis of accelerometer (tremor) and EMG data. Prior to any data analysis, accelerometer and EMG data were filtered using a second-order low-pass Butterworth filter, with cutoff frequencies set at 40 Hz and 400 Hz, respectively. All data analysis was performed using custom software developed in Matlab version 6.0 (MathWorks Release 12).

Time domain analysis of tremor and EMG data consisted of examining the amplitude of the signal obtained from root mean square (RMS) processing of each signal (bin size 100 ms). For the tremor and EMG signals, power spectral analysis was performed on the raw, filtered data within the range of 0–40 Hz and 0–400 Hz, respectively. The power spectral analysis was determined using the SPECTRUM function in Matlab. This analysis estimates the power spectral density of the respective signal using Welch's averaged, modified periodogram method. This analysis was performed using a window size of 512 data points and produced a frequency bin-width for the tremor and EMG signals of 0.1953 Hz and 0.9755 Hz, respectively.

In addition, for the tremor data, separate power spectral analyses were performed within four bandwidths: 0–7, 7–17, 17–30 and 30–40 Hz. These bandwidths were selected since distinct frequency peaks in the tremor profile are typically observed between 2–4, 8–12 Hz and, for the index finger, 18–25 Hz (Elble and Koller 1990; McAuley et al. 1997; Vaillancourt and Newell 2000a; Keogh et al. 2004). The dependent measures calculated for each accelerometer and EMG signal were: peak power, the frequency at which the peak

power was observed [peak power frequency (PPF)], mean peak power (MPP), and the proportion of total power associated with each major peak.

Tremor–tremor, tremor–EMG and EMG–EMG relations were assessed in the time domain using cross-correlation analysis (Pearson product moment) and in the frequency domain using coherence analysis. The peak correlation coefficient, and the respective time lag were used as a measure of the degree of coupling. Cross correlation analysis was performed for each trial using time lags of ± 5 s. As for the power spectral analysis, assessment of the peak coherence between any two signals was calculated using Welch's averaged periodogram method.

EMG–EMG analysis was performed on the original filtered data. Peak coherence measures between the two EMG signals were calculated within the range 0–400 Hz (window size, 512 data points; binwidth = 0.9785 Hz).

For the EMG-tremor cross-correlation and coherence analyses only, the EMG data was down-sampled to 100 Hz prior to analysis since the tremor and EMG signals were sampled at different rates. For tremor–tremor and tremor–EMG relations, peak coherence was assessed within the range 0–40 Hz using the same frequency bandwidths as for the power spectral analysis (window size, 512 data points; binwidth = 0.1953 Hz).

To determine whether the level of coherence between any two signals was significantly different from zero, a 95% confidence interval was calculated according to the methods described by Halliday and colleagues (1995). In addition, assessment of which regions of coherence were significantly different from background coherence levels were determined using the procedures outlined by Vaillancourt and Newell (2000a; 2000b).

Statistical analysis

Inferential analysis of the calculated time and frequency measures was performed using a within-subject, multivariate repeated measures ANOVA design. Subsequent post hoc analyses were performed using honestly significant difference (HSD) test. All tests were performed using SAS statistical software (SAS Institute Inc., Cary, NC, USA), with the risk of Type I error set at $P < 0.01$. All data were reported in text as the group mean $\pm$ one standard error of the mean.

Results

Pre-exercise

Tremor and EMG analysis

For pre-exercise conditions, the average RMS tremor amplitude within limb was greatest at the index finger and decreased systematically from distal to proximal within the upper limb (Exercised limb, Mean finger RMS: 0.221 ± 0.071 m s^{-2}; Mean hand RMS: 0.113 ± 0.028 m s^{-2}; Mean forearm RMS: 0.062 ± 0.016 m s^{-2}; Non-exercised limb, Mean finger RMS: 0.201 ± 0.068 m s^{-2}; Mean hand RMS: 0.111 ± 0.024 m s^{-2}; Mean forearm RMS: 0.056 ± 0.014 m s^{-2}). For all pre-exercise conditions, no significant inter-limb differences were observed in the tremor amplitude (RMS values, peak power) between homologous limb segments, indicating that prior to any exercise intervention, the tremor observed in both limbs was similar in terms of the relative amplitude.

Frequency analysis revealed that the tremor output for all upper limb segments contained at least two prominent components, with the peak power being observed between 2–4 and 8–12 Hz. For the index finger, the majority of power in the tremor output was associated with the 8–12 Hz component (50–57%). In contrast, for the hand and forearm, more power was associated with the 2–4 Hz peak (hand 57–63%; forearm 35–46%). A third, smaller amplitude peak at 18–25 Hz was found in the index finger only. The power associated with this component was always less than 7% of the total power of the tremor signal for the index finger across all conditions. An example of the raw tremor traces and the respective power spectral profiles for both limbs prior to and after the fatiguing exercise are shown in Fig. 1.

Pre-exercise analysis of the EMG activity of the ED muscles (mean RMS, peak power) revealed no significant differences between limbs. For both muscles, the PPF was seen around 120 Hz. (exercised limb PPF = 110 Hz, non-exercised limb PPF = 124 Hz). An example of the level of EMG activity and the respective power spectral profiles for both limbs prior to and after fatiguing exercise is shown in Fig. 2. Analysis of the proportional power data revealed that the majority of the power (55–60%) in the EMG signal for both ED muscles was seen between 50 and 150 Hz. Across all conditions, less than 12% of the power in the EMG signal for either muscle was seen within the range associated with the tremor peaks (below 40 Hz).

Coupling relations

The main result to emerge from the cross correlation and coherence analysis was a significantly high degree of coupling for the tremor in the more distal segments, that is, between the hand and finger. Cross correlation analysis showed that prior to exercise, a significantly high level of coupling in the time domain was observed between the hand and index finger ($r = 0.66$–67, $F_{27,162} = 4.32$, $P < 0.01$). Similar results were attained for the coherence analysis where significantly high values were observed between the tremor for the index finger and hand of both limbs. These significantly higher coherence values were observed within the ranges where both the 2–4 and 8–12 Hz frequency peaks were seen (2–4 Hz peak coherence range 0.91–93,

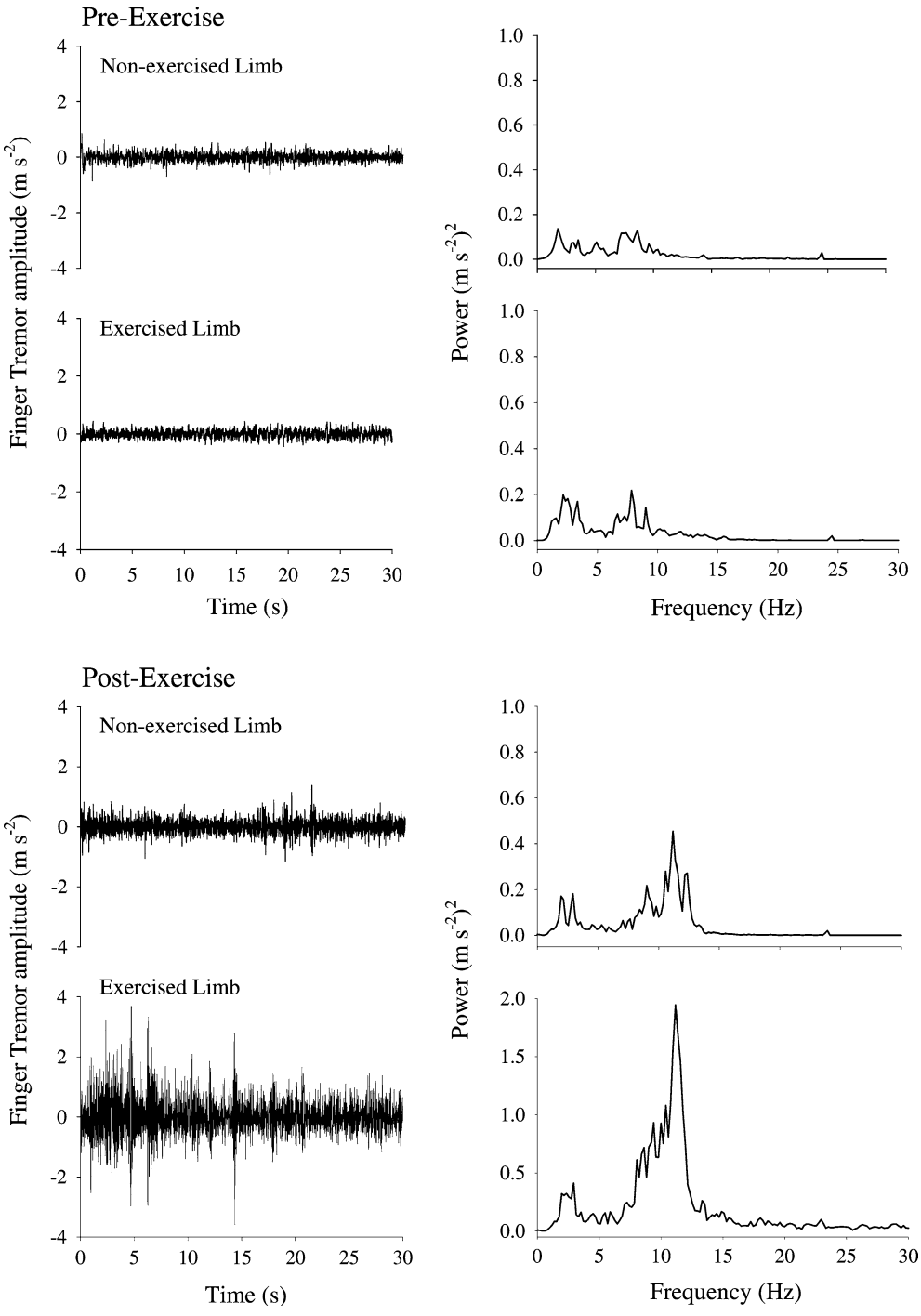

Fig. 1 Representative acceleration (tremor) traces and power spectral density (*PSD*) plots for a single subject pre and post exercise induced fatigue. Tremor traces were obtained from a single trial for the index finger of both the exercised and non-exercised limb. The *left column* contains the postural tremor acceleration for both limbs pre- and post- exercise while the *right hand column* illustrates the respective power spectral plots

$F_{27,\ 162} = 4.37$; 8–12 Hz peak coherence range 0.88–0.90, $F_{27,\ 162} = 6.12$; all P's < 0.01). While a lower degree of coupling was observed between the hand and forearm segments within a single limb (hand-forearm), these relations were non-significant (all r values < 0.30, peak coherence values < 0.40). In addition, no significant relations were observed between the tremor in the index finger and forearm segments of a single limb (all r values < 0.08, peak coherence values < 0.15), or between any contralateral limb segments (all r values < 0.1, peak coherence values < 0.11).

With regard to any tremor–EMG or EMG–EMG relations during the pre-exercise conditions, the results of the time and frequency domain analyses produced no evidence of any significant degree of coupling between the tremor in any limb segment and EMG activity (all r's < 0.05, peak coherence values < 0.18). Nor were there any significant degree of coupling between the ED muscles (all r's < 0.03, peak coherence < 0.28). Examples of cross-correlation and coherence profiles for the specific tremor–tremor, tremor–EMG and EMG–EMG relations are shown in Fig. 3.

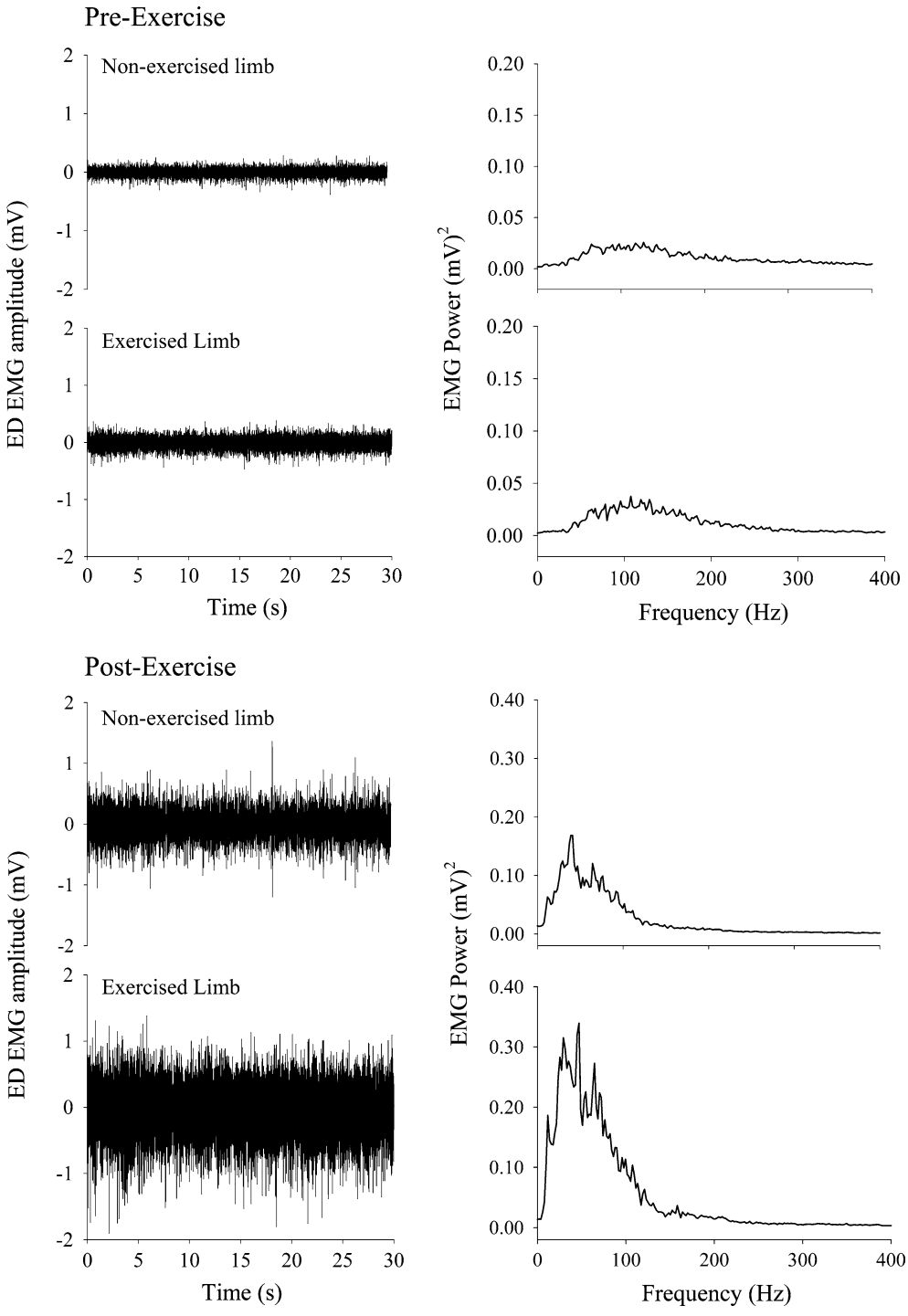

Fig. 2 Representative demodulated EMG traces and PSD plots for the extensor digitorum (ED) muscle for a single subject pre and post exercise induced fatigue. EMG traces were obtained from the ED muscles of both the exercised and non-exercised limb. The *left column* contains the EMG activity for both muscles pre- and post- exercise while the *right hand column* illustrates the respective power spectral plots

Post-exercise

Tremor and EMG analysis

Following fatiguing exercise of the single wrist, an increase in the mean tremor amplitude in all segments of both the exercised and non-exercised limbs was observed. However, the only significant increase was observed for the tremor seen at the index fingers with no significant change in the tremor for the hand and forearm segments of either limb. The results of the RMS analysis showed that the mean tremor amplitude of the index fingers of both limbs increased, irrespective of the fact that only the wrist joint of one limb was exercised ($F_{2,\ 12} = 86.02$; $P < 0.01$). Figure 1 highlighted the general change in the raw tremor and the respective power spectral profiles after the fatiguing exercise. Figure 4 illustrates the changes seen

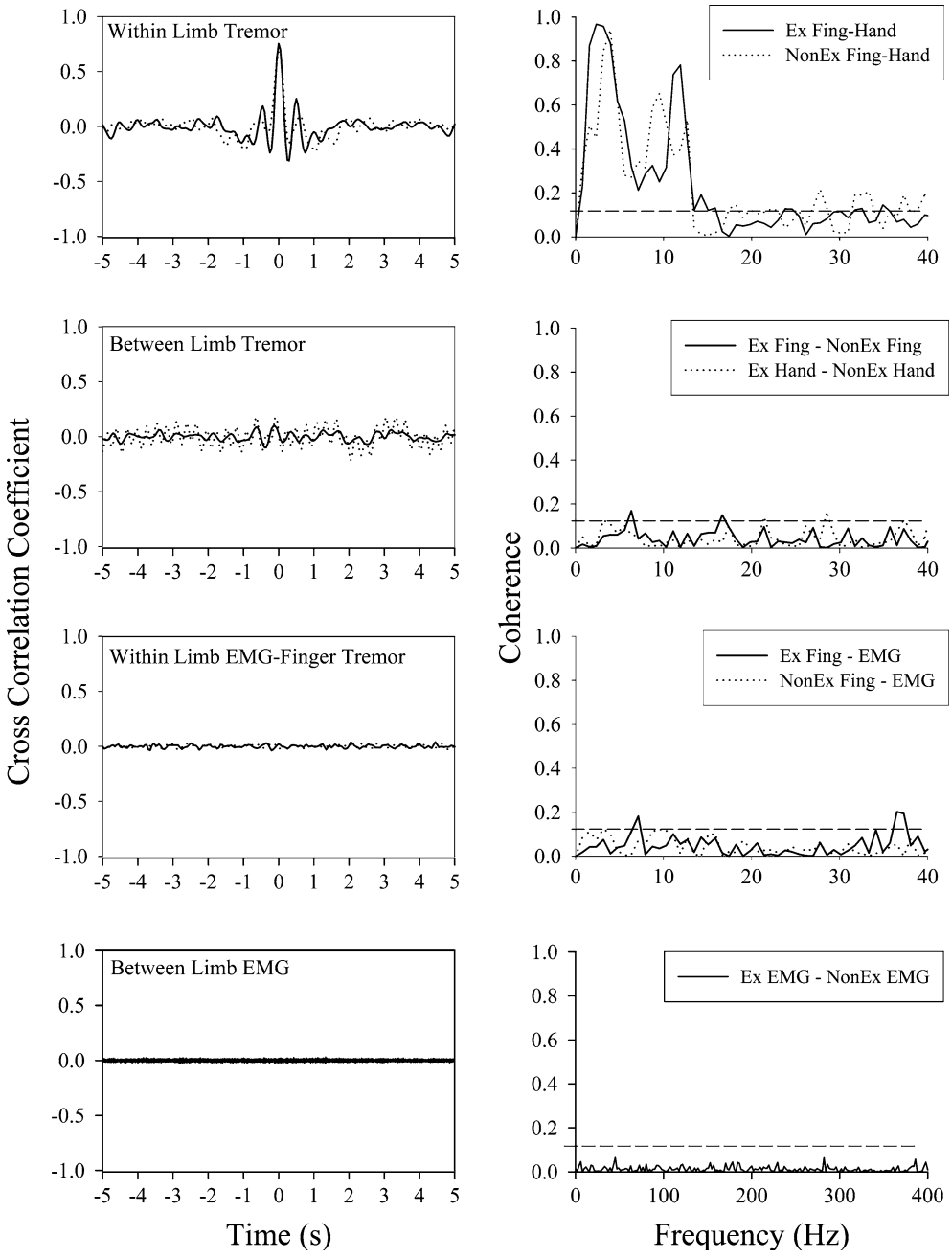

Fig. 3 Cross correlation and coherence plots. The respective coupling relations for the tremor within limb (*hand–finger*) and between limb (*finger–finger*, *hand–hand*) are shown. EMG-finger tremor relations are also presented. All cross correlation plots are shown with time lags to ±5 s. The cutoff for a 95% CI for a coherence level significantly different from 0 (0.125) is shown

in the mean RMS index finger tremor of both limbs and the mean RMS EMG activity for the ED muscles of both limbs as a consequence of the fatiguing exercise.

A significant change in the resultant frequency profile of the index finger tremor for both the exercised and non-exercised arms was also seen following exercise-induced fatigue of the wrist extensors of a single limb. For the index fingers of both limbs, significant increases in the peak power of the 8–12 Hz tremor component ($F_{2,\ 12} = 18.27$; $P < 0.01$) and the proportional power associated with the 8–12 Hz component ($F_{2,\ 12} = 7.83$; $P < 0.01$) were seen. In general, the increase in proportional power associated with the 8–12 Hz peak was of the order of 8–10%. No significant change in the peak power of the 2–4 Hz component was observed for the finger segments, nor was any significant change seen in the peak power of the hand and forearm segments in either the exercised or non-exercised limb following fatigue. Figure 1 provides an example of the raw tremor traces and the respective power spectral profiles for both the exercise and non-exercised limb after the fatiguing exercise.

Fatiguing exercise also produced a significant change on the mean RMS EMG amplitude, where a significant increase in the level of activity in both ED muscles

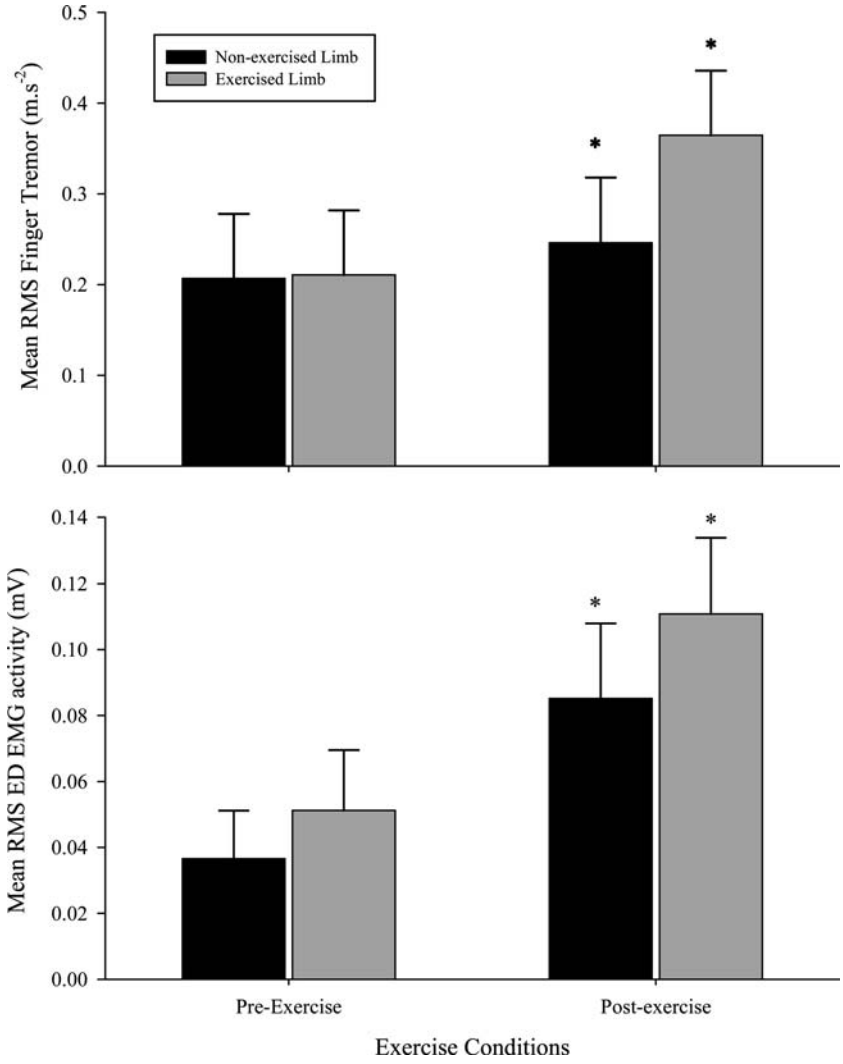

Fig. 4 Changes seen in the mean RMS index finger tremor amplitude and RMS EMG activity for the ED muscles of the exercised and non-exercised limbs prior to and after exercise-induced fatigue. Each value is based upon average values for all subjects ($n = 7$) and across all trials ($n = 8$). Error bars represent one standard error of the mean, above and below each group mean error point. Significant differences between values are denoted with an asterisk

($F_{2,\ 12} = 46.90$, $P < 0.01$) was observed between pre- and post-exercise conditions. Subsequent analysis revealed that the greatest overall increase in the mean RMS amplitude of EMG activity was observed for the exercised limb. In addition, a significant between limb effect was also seen, with a greater level of EMG activity being observed for the ED muscle of the exercised limb as compared to the non-exercised limb ($F_{1,\ 6} = 54.23$, $P < 0.01$). This effect was preserved across all conditions.

The results of the frequency analysis showed that the MPP and the peak power of the EMG activity in both muscles increased significantly following fatigue of a single limb (exercised limb ED, $F_{2,\ 12} = 32.17$; non-exercised limb ED, $F_{2,\ 12} = 27.64$, all P's < 0.01). There was also a resultant change in the frequency value at which the peak power was observed. Following fatiguing exercise, there was a significant shift in the peak power of the ED muscle of both the exercised limb and the non-exercised limb towards lower frequencies (exercised limb PPF = 73 Hz, $F_{2,\ 12} = 9.73$: non-exercised limb PPF = 81 Hz; $F_{2,\ 12} = 6.21$, all P's < 0.01). The relative changes in the level of EMG activity and the respective power spectral profiles for both limbs following exercise was previously shown in Fig. 2. Fatiguing the muscles about the wrist joint of a single arm also resulted a significant shift in the proportional power of the EMG output, with more of the EMG signal (62–68%) for both ED muscles being seen between 50 and 150 Hz (exercised limb $F_{2,\ 12} = 5.96$: non-exercised limb $F_{2,\ 12} = 4.13$, all P's < 0.01). Figure 5 highlights the relative changes in the proportional power for the index finger tremor of both limbs and the EMG signals for both ED muscles as a consequence of exercise-induced fatigue.

Lactate

Blood lactate measures were taken from each subject at four different time periods during the experiment. Two tests were taken during the pre-exercise testing: (1) at rest, prior to data collection, and (2) immediately following pre-exercise trials. No significant differences were

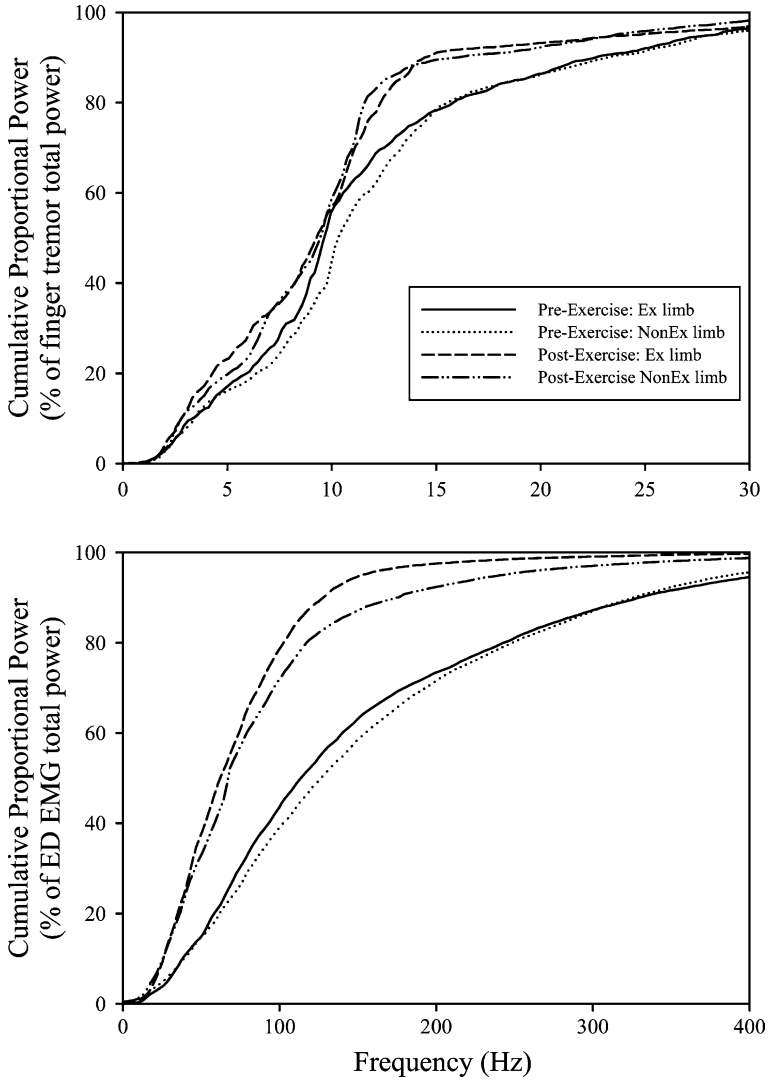

Fig. 5 Ensemble averages of cumulative changes in proportional power for the ED EMG activity (*bottom*) and index finger tremor (*top*). Each trace highlights the differences in proportional power for the tremor and EMG of the exercised and non-exercised limbs prior to and after exercise-induced fatigue

observed in lactate levels between the two different time periods prior to exercise (mean lactate values exercised limb, time (1) 1.670 $mmol^{-1}$, time (2) 1.671 $mmol^{-1}$; non-exercised limb time (1) 1.628 $mmol^{-1}$, time (2) 1.614 $mmol^{-1}$).

Following exercise, two further blood lactate measures were taken from each subject. These samples were taken: (3) immediately following exercise, and (4) immediately following post-exercise trials. Immediately following exercise, there was a significant increase in lactate values for the exercised limb (mean lactate value 2.15 $mmol^{-1}$; $F_{2,\ 12} = 7.71$; $P < 0.01$) but no significant difference in the lactate values for the non-fatigued limb as compared to the pre-exercise levels (mean lactate value 1.682 $mmol^{-1}$). The lactate values attained immediately following post-exercise trials (time 4) demonstrated no significant difference between limbs or in comparison to the pre-exercise levels (mean lactate values, exercised limb 1.82 $mmol^{-1}$; non-exercised limb 1.617 $mmol^{-1}$). Figure 6 illustrates the relative changes in blood lactate levels of each limb as a function of the different time periods before and after exercise. Additionally, the mean tremor RMS amplitude values for each trial are shown to illustrate the relative changes in tremor output in relation to changes in lactate levels. Although non-significant, under post-exercise conditions a trend is evident where a decrease in lactate level corresponds to an increase in tremor in both limbs.

Coupling relations

Following fatiguing exercise, the only significant change seen in the coupling relations was seen in the cross correlation values for the hand-finger combination of the exercised limb. For this combination, the strength of the hand-finger correlations decreased significantly (exercised limb $r = 0.33$–0.42; $F_{27,\ 162} = 6.10$, $P < 0.01$), but no change was observed in the hand-finger relations for the non-exercised limb.

Despite the change in the cross correlation values, no significant change in the peak coherence values between

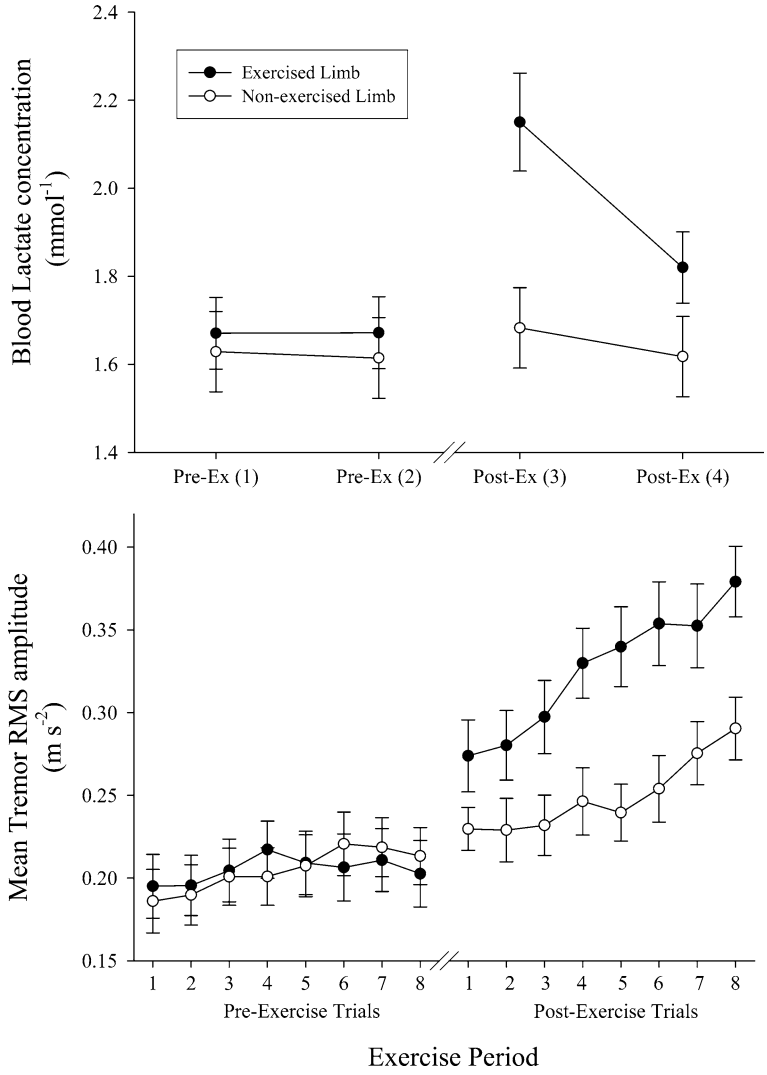

Fig. 6 Changes in mean lactate and mean RMS finger tremor outputs both prior to and following exercise-induced fatigue. Mean RMS tremor values for the index fingers of each limb represent group averages for each trial performed. *Error bars* represent one standard error of the mean, above and below each group mean error point

the hand and finger for either limb were seen following exercise. In addition, exercise was shown to result in no significant change in the level of coupling between the tremor in any limb segment and EMG activity (all r's <0.04, peak coherence values < 0.28) or on the degree of coupling between the ED muscles (all r's < 0.03, peak coherence <0.28).

Discussion

The aim of this study was to examine the post-exercise effects of fatiguing the wrist extensor muscles of a single arm on postural tremor and muscle activity in both arms. The major finding of this study was that, following the unilateral fatiguing protocol, an increase in the amplitude of physiological tremor and EMG activity in both limbs was observed. For tremor, this effect was confined to the finger segments only with no significant change in the amplitude of tremor for the hand and forearm.

Bilateral effects of unilateral fatigue

Following exercise-induced fatigue of the wrist extensors of a single arm, an increase in forearm muscle activity and index finger tremor for both limbs was seen. These increases in amplitude were reflected by changes in both the time (RMS amplitude) and frequency (peak power, proportional power, MPP) measures of the EMG and tremor. Interestingly, the increases in tremor amplitude were confined to the index fingers, with no significant increase in the tremor for the hand and forearm of either arm following unilateral exercise. In addition, these bilateral increases in physiological tremor and EMG activity of ED were only observed during the bilateral pointing task, with no changes in tremor or EMG activity seen for the non-exercised limb during the unilateral exercise. While previous research has shown that localised fatigue does produce a resultant increase in the amplitude of postural tremor within the fatigued limb (Bousfield 1932; Lippold 1981; Palmer 1991; Gajewski and Viitasalo 1994; Saxton et al. 1995), the finding of

bilateral increases in the finger tremor and EMG activity following unilateral fatigue has not been reported to date.

Frequency analysis revealed that the effect of fatigue on bilateral finger tremor was confined to an increase in the peak power of the 8–12 Hz peak. Across all conditions, the tremor observed within any single segment of the upper limb was characterised by at least two prominent frequency peaks, as previously reported (Morrison and Newell 1996; Morrison and Newell 1999; Keogh et al. 2004). These peaks were observed between the 2–4 and 8–12 Hz for the forearm, hand and finger segments while a third, higher frequency peak (between 18–25 Hz) was also seen in the index finger. While unilateral fatigue resulted in a bilateral increase in the peak power and proportional power of the 8–12 Hz tremor component of the finger tremor, no significant changes in the other finger tremor peaks (2–4 and 18–25 Hz) were observed following the exercise intervention.

The 8–12 Hz tremor peak seen for the index finger is believed to represent the combined output from such central neural structures as the inferior olive, basal ganglia, thalamus (Llinas 1984; Marsden 1984; Elble and Koller 1990; Elble 1996; Llinas and Pare 1997) and, at the spinal cord, the alpha motorneurons (Elble and Randall 1976; Stiles 1980; Elble and Koller 1990; McAuley et al. 1997; Halliday et al. 1999). The fact that a bilateral increase in 8–12 Hz power and EMG activity was seen post-exercise would indicate that an increase in the central neural output is probably responsible for this change. While there is no doubt that peripheral neural factors such as the stretch reflex play a role in the generation of tremor, its contribution is typically associated with the mechanical resonant component of the segment tremor (Stiles and Randall 1967; Lippold 1970; Stiles and Pozos 1976; Elble and Randall 1978; Stiles 1980; Elble and Koller 1990; Vaillancourt and Newell 2000). As no change in the resonant component of the index finger (between 18–25 Hz) was seen, it seems likely that the increases seen in tremor following exercise were not mediated by changes in reflex activity. However, as the output of the muscle spindles and other mechanoreceptors can be altered during fatigue (Gandevia 1998; Taylor et al. 2000; Gandevia 2001), the possibility that changes in the afferent drive from the exercised limb may contribute to the contralateral increase in tremor EMG activity cannot be completely discounted.

In addition to observing changes in tremor activity, increases in amplitude of EMG activity (RMS amplitude, peak power and MPP) and a change in the frequency profile (PPF) of the EMG output were also observed. As with the tremor changes, these increases were seen for the ED muscles of both limbs following exercise of a single arm. Although changes in spectral properties are typically associated with the exercised muscle group (De Luca 1997), it is worth noting that there was also an increase in peak power, MPP and the shift in the PPF of the EMG signal for the extensor muscle of the non-exercised limb. Interestingly, no significant change in EMG activity within the non-exercised limb was seen during the actual exercise protocol. This result indicates that the bilateral changes in EMG activity seen post-exercise were directly associated with performing the pointing task.

The results showed that the effects of fatigue were confined to the finger tremor of both limbs, with no alteration of the tremor in the hand or forearm of either arm. While bilateral changes in tremor amplitude have been reported following perturbation of a single limb (Morrison and Newell 1999), no contralateral changes in tremor following fatigue have been previously demonstrated. Similarly, various studies have reported increased EMG activity in muscles located contralateral to the target muscle during tasks which require a moderate-to-high force output following fatigue, (Zijdewind et al. 1990; Taylor et al. 2000; Zijdewind and Kernell 2001). However, these bilateral effects tend to be observed when the force levels exceeds 30% MVC (Zijdewind et al. 1998). What the current study highlights is that a bilateral effect was seen during the performance of a postural task with low force requirements following exercise of a single arm. This apparent disparity in the bilateral effects following unilateral exercise supports the general view that the impacts of fatigue are heavily influenced by the nature of the task (Enoka 1995; Gandevia 2001).

As both the 8–12 Hz tremor and EMG events are neurally derived, it seems likely that the bilateral changes observed were the result of an increase in the central neural drive to both limbs. However, a relevant question to be asked is what contribution does peripheral fatigue have on the bilateral tremor and EMG changes observed? With regard to the markers of peripheral fatigue, immediately post-exercise, there was an increase in the blood lactate levels for the exercised limb and no significant change in the lactate levels for the non-exercised limb. While the elevated lactate values in the exercised limb declined quickly to pre-exercise levels following fatigue, there was a corresponding increase in the mean RMS amplitude of finger tremor in both limbs over the eight post-exercise trials (see Fig. 6). This increase was only seen for finger tremor, with no increase in forearm and hand tremor following fatigue-inducing exercise. The progressive bilateral increases in finger tremor and EMG activity post-exercise combined with the reciprocal decrease in lactate levels for the exercised limb over time, would suggest that the contralateral effects were probably derived from changes in the neural output rather than increases in local metabolites due to fatigue. Although singular changes in lactate levels are not the only indicator of peripheral fatigue, any significant alteration in their levels has generally been shown to provide an indicator of peripheral fatigue (Vollestad et al. 1988; Kent-Braun 1999; Todd et al. 2003; Gladden 2004).

The results of the coupling analysis provide further evidence to support the view that the bilateral changes are driven by a change in the neural drive to both

limbs. Under pre-exercise conditions, there was no coupling of tremor or EMG activity between limbs, as evidenced by the cross correlation and coherence results. The lack of any inter-limb tremor or tremor–EMG relations is consistent with prior research (Marsden et al. 1969; Morrison and Newell 1996; Morrison and Newell 1999; Keogh et al. 2004). Following the exercise, there was no significant change in the inter-limb tremor or EMG–EMG relations. Thus while subjects were still able to preserve a degree of independence between limbs, the effects of fatiguing the wrist of a single arm were still seen bilaterally. It would seem that in the act of trying to minimise tremor in the exercised limb, increased activation of the homologous muscles of both limbs occurred. A similar bilateral change in postural tremor with no alteration in inter-limb coupling has been reported under conditions where the range of motion of a single arm was perturbed by applying splints across the joints of the arm (Morrison and Newell 1999).

Tremor–EMG relations

The effects of fatigue were primarily confined to changes in EMG and tremor amplitude, with no significant change in the degree of coupling between muscle activity and tremor (in any segment) after exercise. While this result is consistent with previous multiple segment tremor research (Keogh et al. 2004), it contrasts with the reports of other tremor studies who have reported significant relations between physiological tremor and muscle activity (McAuley et al. 1997; Vaillancourt and Newell 2000a; Vaillancourt et al. 2003). The most likely explanation for these differences is that the tremor seen in any segment is highly dependent on the actual task being performed (Keogh et al. 2004). For example, McAuley et al (1997), Vaillancourt and Newell (2000a) and Vaillancourt et al. (2003) all examined tremor–EMG relations within a single segment and during postural tasks where the hand and forearm were supported. As no external support of the hand or finger was employed in the current study, the act of holding the entire arm against gravity was reflected in an increased level of EMG activity in comparison to single segment studies, and a frequency distribution more typical of multiple segment actions. For this postural task, the majority of the power in the EMG signal was seen between 50 and 150 Hz. Less than 12% of the power in the EMG signal for either muscle was seen below 40 Hz, which was the range where the majority of the power in the finger tremor signal (98%) was seen. In contrast, the majority of the power in the EMG frequency distribution reported by McAuley et al. (1997), Vaillancourt and Newell (2000a) and Vaillancourt et al. (2003) was generally observed below 50 Hz. The reported differences in the EMG frequency profile during single and multiple segment tasks, probably highlights the fact that any tremor–EMG relation is influenced by factors such as the number of segments involved and the task being performed.

Conclusion

Overall the results show that fatiguing the muscles about a distal segment of a single arm resulted in an increase in finger tremor and EMG activity in both arms following the exercise. The resultant bilateral increases seen in EMG activity and the neurally generated 8–12 Hz tremor components indicate that these contralateral changes were neurally mediated. Together these results challenge the general assumption of bilateral independence of tremor production and also provide further evidence as to the task specific nature of fatigue.

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